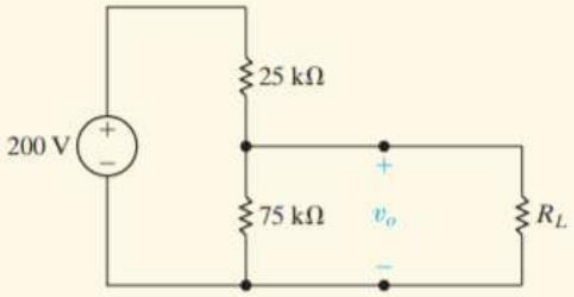
ELEC221 Week 4 Tutorial (Fall 2023)

3.2 a) Find the no-load value of v_o in the circuit shown.

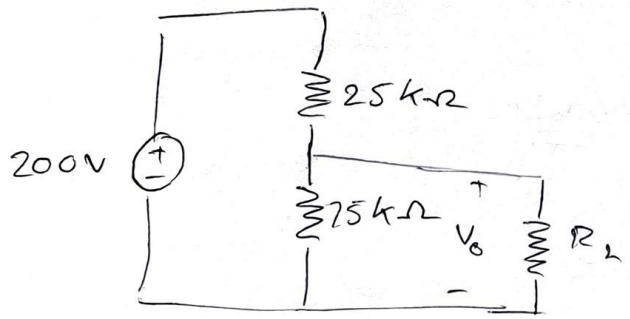
b) Find v_o when R_L is 150 kΩ.

c) How much power is dissipated in the 25 kΩ resistor if the load terminals are accidentally short-circuited?

d) What is the maximum power dissipated in the 75 kΩ resistor?



Answer: (a) 150 V;
(b) 133.33 V;
(c) 1.6 W;
(d) 0.3 W.



a) ignore R_2, treat terminals as open circuit

$$200\,V = i(25\,k\Omega) + i(75\,k\Omega)$$

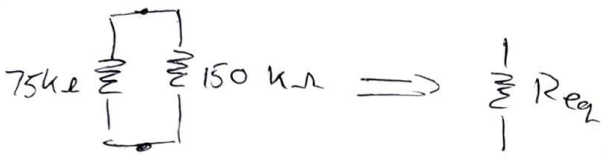
$$i = 2\,mA$$

$$V_o = (2\,mA)(75\,k\Omega) = \underline{\underline{150\,V}}$$

or

$$V_o = (200\,V)\left(\frac{75\,k\Omega}{25\,k\Omega + 75\,k\Omega}\right) = \underline{\underline{150\,V}}$$

b)



$$R_{eq} = \left(\frac{1}{75k} + \frac{1}{150k}\right)^{-1} = 50\,k\Omega$$

$$V_o = (200\,V)\left(\frac{50\,k\Omega}{25\,k\Omega + 50\,k\Omega}\right) = \underline{133.3\,V}$$

c) $P = \frac{V^2}{R} = \frac{200^2}{25\,k\Omega} = \underline{1.6\,W}$

or

$$i = \frac{200\,V}{25\,k\Omega} = 8\,mA$$

$$P = i^2R = iV = (8\,mA)^2(25\,k\Omega) = (200\,V)(8\,mA) = 1.6\,W$$

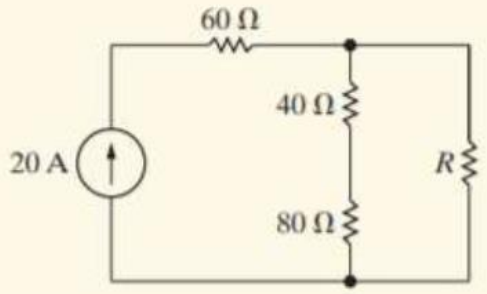
d) Power dissipated in 75 kΩ resister is at maximum when at no-load because V_o is at its maximum

$$P_{75k\,max} = \frac{V^2}{R} = \frac{150^2}{75\,k} = \underline{0.3\,W}$$

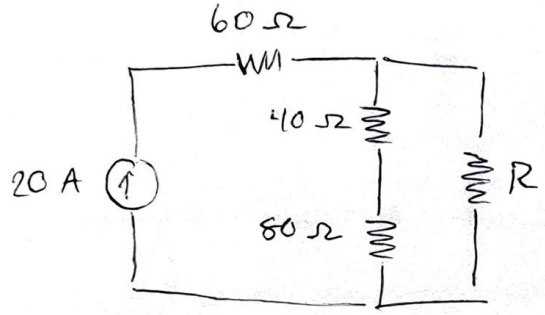
3.3 a) Find the value of R that will cause 4 A of current to flow through the 80 Ω resistor in the circuit shown.

b) How much power will the resistor R from part (a) need to dissipate?

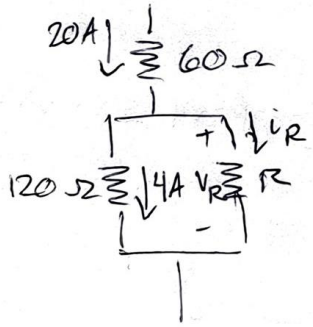
c) How much power will the current source generate for the value of R from part (a)?



Answer: (a) 30 Ω;
(b) 7680 W;
(c) 33,600 W.



a) combining series resistors we get



By KCL, $i_R = 16A$

Since $i_R = 4 \cdot (4A)$, $R = \frac{1}{4}(120\Omega)$

so $R = \underline{30\Omega}$

or

By KVL, $V_R = (4A)(120\Omega) = 480V$

$i_R = \frac{V_R}{R} \Rightarrow 16A = \frac{480V}{R} \Rightarrow R = \underline{30\Omega}$

b) $P_R = i_R^2 R = (16A)^2(30\Omega) = \underline{7.68kW}$

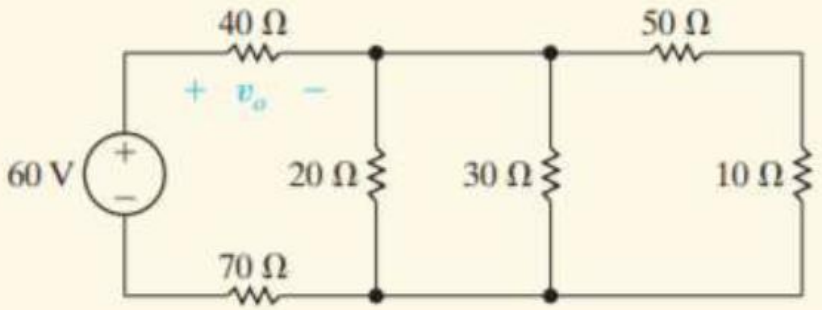
c) $V_{source} = V_{60\Omega} + V_R = (20A)(60\Omega) + 480 = 1680V$

$P_{source} = iV = (20A)(1680V) = \underline{33.6kW}$

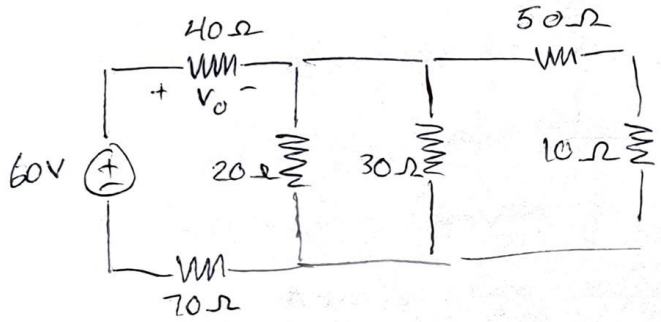
3.4 a) Use voltage division to determine the voltage v_o across the 40 Ω resistor in the circuit shown.

b) Use v_o from part (a) to determine the current through the 40 Ω resistor, and use this current and current division to calculate the current in the 30 Ω resistor.

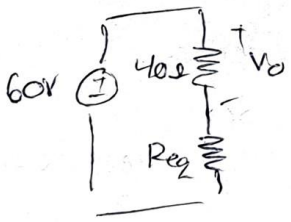
c) How much power is absorbed by the 50 Ω resistor?



Answer: (a) 20 V;
(b) 166.67 mA;
(c) 347.22 mW.



a) Simplify to



$R_{eq} = [20 \| 30 \| (50+10)] + 70\,\Omega$

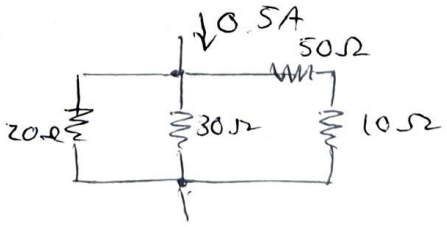
$R_{eq} = \left(\frac{1}{20} + \frac{1}{30} + \frac{1}{60}\right)^{-1} + 70\,\Omega$

$R_{eq} = 10\,\Omega + 70\,\Omega = 80\,\Omega$

$V_o = (60\,V)\left(\frac{40\,\Omega}{80\,\Omega + 40\,\Omega}\right)$

$V_o = \underline{20\,V}$

b) $i_{40\Omega} = \frac{20V}{40\,\Omega} = \underline{0.5A}$



Combine 20Ω, 50Ω, and 10Ω into R_{eq}.

$R_{eq} = 20 \| (50+10) = 15\,\Omega$

$i_{30\Omega} = (0.5A)\left(\frac{15\,\Omega}{15\Omega + 30\,\Omega}\right) = \underline{0.1667A}$

c) First find $i_{50\Omega}$

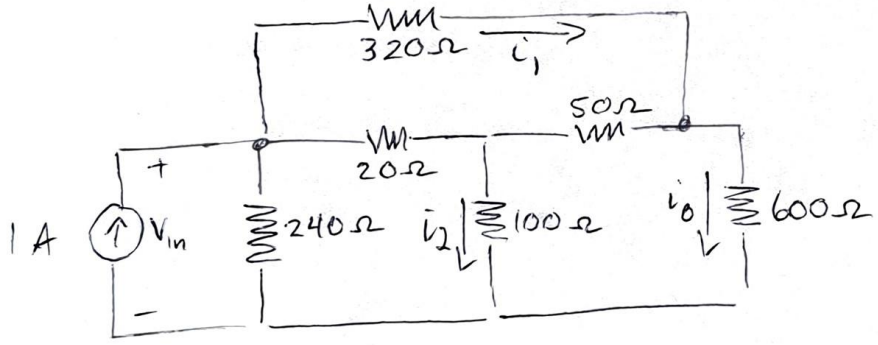
$R_{eq} = 20 \| 30 = 12\,\Omega$

$i_{50\Omega} = (0.5A)\left(\frac{12\,\Omega}{12\Omega + 60\,\Omega}\right) = 0.0833A$

$P_{50\Omega} = (0.0833)^2(50) = 347.22\,mW$

Solution, Problem 4,

Use a Y-Δ transformation to find a) i_0, b) i_1, c) i_2, and d) the power developed by the current source



Answers:

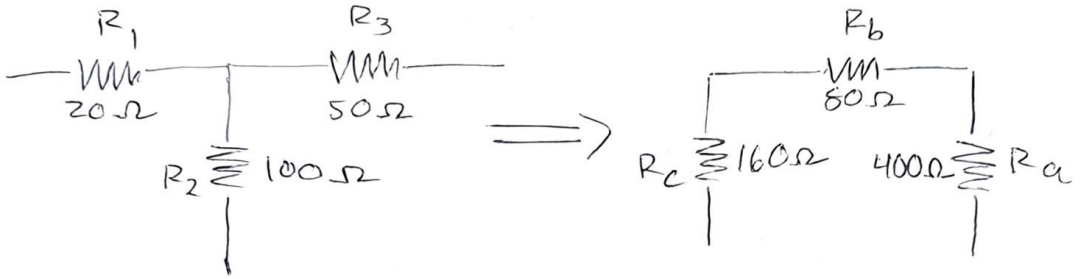
$i_0 = 0.096\,A$

$i_1 = 0.048\,A$

$i_2 = 0.6\,A$

$P_{source} = -72.96\,W$ or $72.96\,W$ of power delivered

First, do the Y-Δ transformation

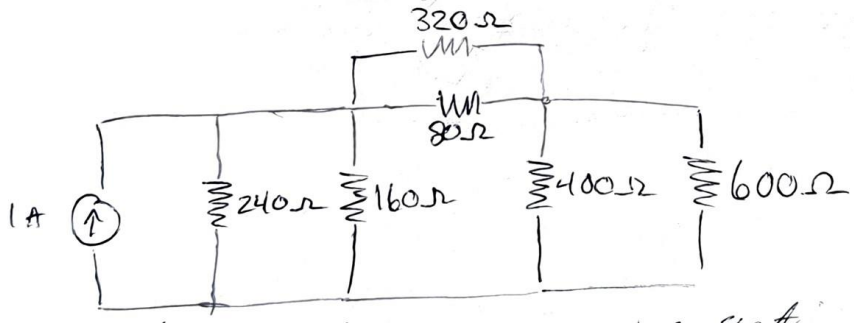


$$R_a = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_1} = 400\,\Omega$$

$$R_b = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_2} = 80\,\Omega$$

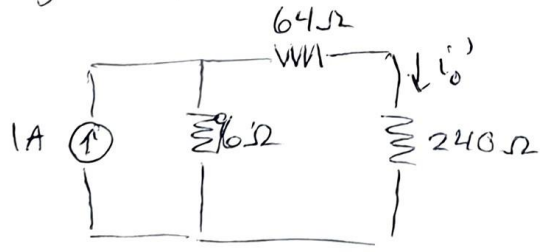
$$R_c = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_3} = 160\,\Omega$$

So now we have the equivalent circuit



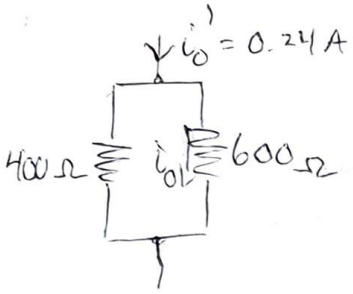
$600 \| 400 = 240\,\Omega$
$80 \| 320 = 64\,\Omega$
$240 \| 160 = 96\,\Omega$

Combining the parallel resistances, we get:

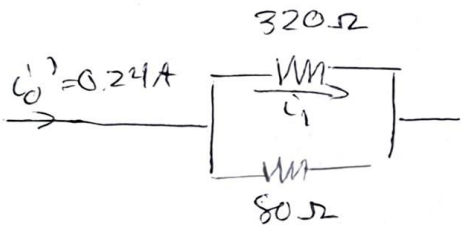


$$i_0' = (1A)\left(\frac{96}{96 + (64+240)}\right)$$

$$i_0' = 0.24A$$

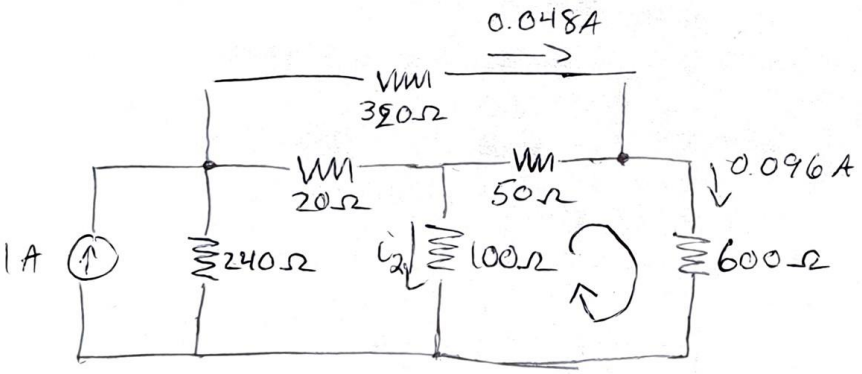


so $i_o = i_o'\left(\frac{400}{400+600}\right) = \underline{0.096A}$



so $i_1 = i_o'\left(\frac{80}{80+320}\right) = \underline{0.048A}$

To find i_2, we must examine the original circuit.



By KCL, $i_{50\Omega} = (0.096A - 0.048A) = 0.048\ A$

Now apply KVL around the 100Ω-50Ω-600Ω loop.

$$V_{100\Omega} = V_{50\Omega} + V_{600\Omega}$$

$$i_2(100\Omega) = (0.048A)(50\Omega) + (0.096A)(600\Omega)$$

$$i_2 = \underline{0.6A}$$

By KVL

$$V_{in} = V_{240\Omega} = V_{20\Omega} + V_{100\Omega}$$

From KVL, $i_{20\Omega} = i_2 + i_{50\Omega} = (0.6A + 0.048A) = 0.648A$

so $V_{in} = i_{20\Omega}(20\Omega) + i_2(100\Omega) = 72.96V$

$$P_{source} = -V_{in}\, i_{in} = -(72.96\,V)(1A) = -72.96\,W$$

The source delivers 72.96 W of power